LAB ASSIGNMENT – SQL FUNCTIONS

APPLY SQL FUNCTIONS: STRING, NUMERIC, DATE, AND AGGREGATE FUNCTIONS

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1. AIM

To apply and demonstrate SQL String, Numeric, Date, and Aggregate Functions using a School Student Database.

2. OBJECTIVE

The objectives of this assignment are:

1. To understand the concept of SQL Functions.
2. To apply String Functions for manipulating text data.
3. To apply Numeric Functions for performing calculations.
4. To apply Date Functions for working with date values.
5. To apply Aggregate Functions for performing calculations on multiple rows.
6. To use the GROUP BY clause with aggregate functions.
7. To combine multiple SQL functions in a single query.

SQL functions accept input, perform an operation, and return an output.

3. DATABASE SCENARIO

A school database is used to maintain basic information about its students.

The database contains a table named **STUDENT** with information such as Student ID, Student Name, Class, Marks, Fee Paid, Date of Birth, and Admission Date.

4. CREATION OF STUDENT TABLE

INSERTING RECORDS INTO STUDENT TABLE

DISPLAYING THE STUDENT TABLE

```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
```

```
STDIN
Input for the program ( Optional )

Output:
+------------+------------------+-------+-------+----------+---------------+----------------+
| Student_ID | Student_Name     | Class | Marks | Fee_Paid | Date_of_Birth | Admission_Date |
+------------+------------------+-------+-------+----------+---------------+----------------+
|        101 | Aditi M          | 10A   |    85 | 50000.00 | 2008-05-15    | 2023-06-10     |
|        102 | Bhavya Sahithi G | 10A   |    88 | 50000.00 | 2008-11-20    | 2023-06-12     |
|        103 | Geetha S         | 10B   |    76 | 45000.00 | 2009-02-10    | 2023-06-11     |
|        104 | Harshitha J      | 10B   |    92 | 48000.00 | 2008-08-25    | 2023-06-15     |
|        105 | Kirthana K       | 10A   |    70 | 40000.00 | 2009-01-30    | 2023-06-13     |
+------------+------------------+-------+-------+----------+---------------+----------------+
```

PART A – STRING FUNCTIONS

String functions are used to manipulate text data. In this table, Student_Name and Class are examples of text data

UPPER() FUNCTION

Purpose

The UPPER() function converts text into uppercase letters

```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;
```

```
STDIN
Input for the program ( Optional )

Output:
+------------+------------------+-------+-------+----------+---------------+----------------+
| Student_ID | Student_Name     | Class | Marks | Fee_Paid | Date_of_Birth | Admission_Date |
+------------+------------------+-------+-------+----------+---------------+----------------+
|        101 | Aditi M          | 10A   |    85 | 50000.00 | 2008-05-15    | 2023-06-10     |
|        102 | Bhavya Sahithi G | 10A   |    88 | 50000.00 | 2008-11-20    | 2023-06-12     |
|        103 | Geetha S         | 10B   |    76 | 45000.00 | 2009-02-10    | 2023-06-11     |
|        104 | Harshitha J      | 10B   |    92 | 48000.00 | 2008-08-25    | 2023-06-15     |
|        105 | Kirthana K       | 10A   |    70 | 40000.00 | 2009-01-30    | 2023-06-13     |
+------------+------------------+-------+-------+----------+---------------+----------------+
+------------------+------------------+
| Student_Name     | Uppercase_Name   |
+------------------+------------------+
| Aditi M          | ADITI M          |
| Bhavya Sahithi G | BHAVYA SAHITHI G |
| Geetha S         | GEETHA S         |
| Harshitha J      | HARSHITHA J      |
| Kirthana K       | KIRTHANA K       |
+------------------+------------------+
+------------------+------------------+
| Student_Name     | Lowercase_Name   |
+------------------+------------------+
| Aditi M          | aditi m          |
| Bhavya Sahithi G | bhavya sahithi g |
| Geetha S         | geetha s         |
| Harshitha J      | harshitha j      |
| Kirthana K       | kirthana k       |
+------------------+------------------+
```

LOWER() FUNCTION

Purpose

The LOWER() function converts text into lowercase letters.

```sql
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;
```

STDIN
Input for the program (Optional)

Output:

Student_ID	Student_Name	Class	Marks	Fee_Paid	Date_of_Birth	Admission_Date
101	Aditi M	10A	85	50000.00	2008-05-15	2023-06-10
102	Bhavya Sahithi G	10A	88	50000.00	2008-11-20	2023-06-12
103	Geetha S	10B	76	45000.00	2009-02-10	2023-06-11
104	Harshitha J	10B	92	48000.00	2008-08-25	2023-06-15
105	Kirthana K	10A	70	40000.00	2009-01-30	2023-06-13

Student_Name	Uppercase_Name
Aditi M	ADITI M
Bhavya Sahithi G	BHAVYA SAHITHI G
Geetha S	GEETHA S
Harshitha J	HARSHITHA J
Kirthana K	KIRTHANA K

Student_Name	Lowercase_Name
Aditi M	aditi m
Bhavya Sahithi G	bhavya sahithi g
Geetha S	geetha s
Harshitha J	harshitha j
Kirthana K	kirthana k

LENGTH() FUNCTION

Purpose

The LENGTH() function finds the number of characters in a string.

```sql
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;
```

STDIN
Input for the program (Optional)

Output:

Student_ID	Student_Name	Class	Marks	Fee_Paid	Date_of_Birth	Admission_Date
101	Aditi M	10A	85	50000.00	2008-05-15	2023-06-10
102	Bhavya Sahithi G	10A	88	50000.00	2008-11-20	2023-06-12
103	Geetha S	10B	76	45000.00	2009-02-10	2023-06-11
104	Harshitha J	10B	92	48000.00	2008-08-25	2023-06-15
105	Kirthana K	10A	70	40000.00	2009-01-30	2023-06-13

Student_Name	Uppercase_Name
Aditi M	ADITI M
Bhavya Sahithi G	BHAVYA SAHITHI G
Geetha S	GEETHA S
Harshitha J	HARSHITHA J
Kirthana K	KIRTHANA K

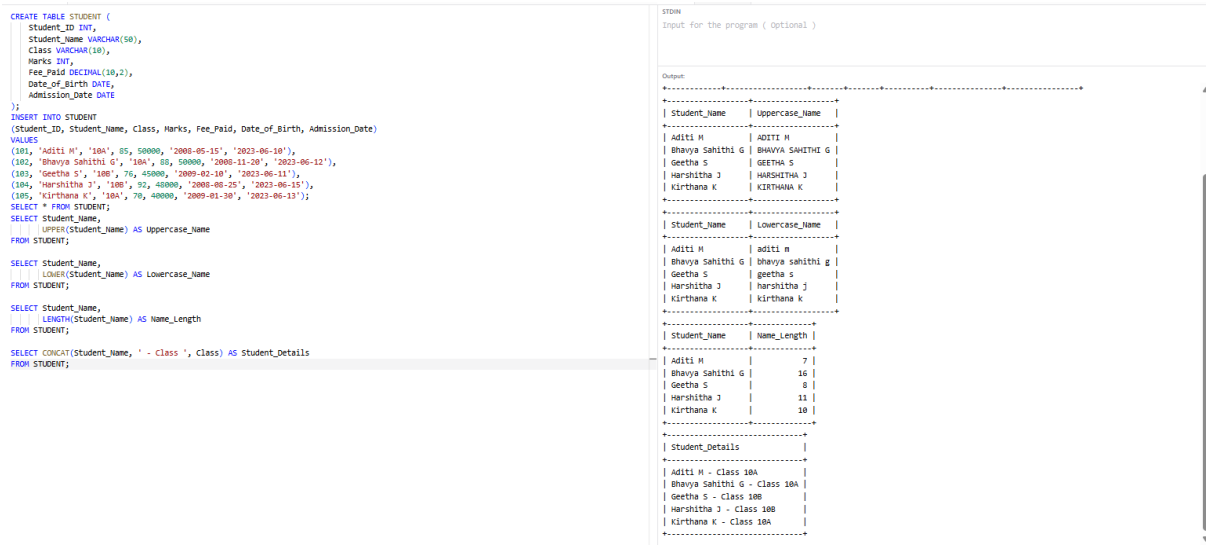
Student_Name	Lowercase_Name
Aditi M	aditi m
Bhavya Sahithi G	bhavya sahithi g
Geetha S	geetha s
Harshitha J	harshitha j
Kirthana K	kirthana k

Student_Name	Name_Length
Aditi M	7
Bhavya Sahithi G	16
Geetha S	8
Harshitha J	11
Kirthana K	10

CONCAT() FUNCTION

Purpose

The CONCAT() function joins two or more strings together.



```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;
```

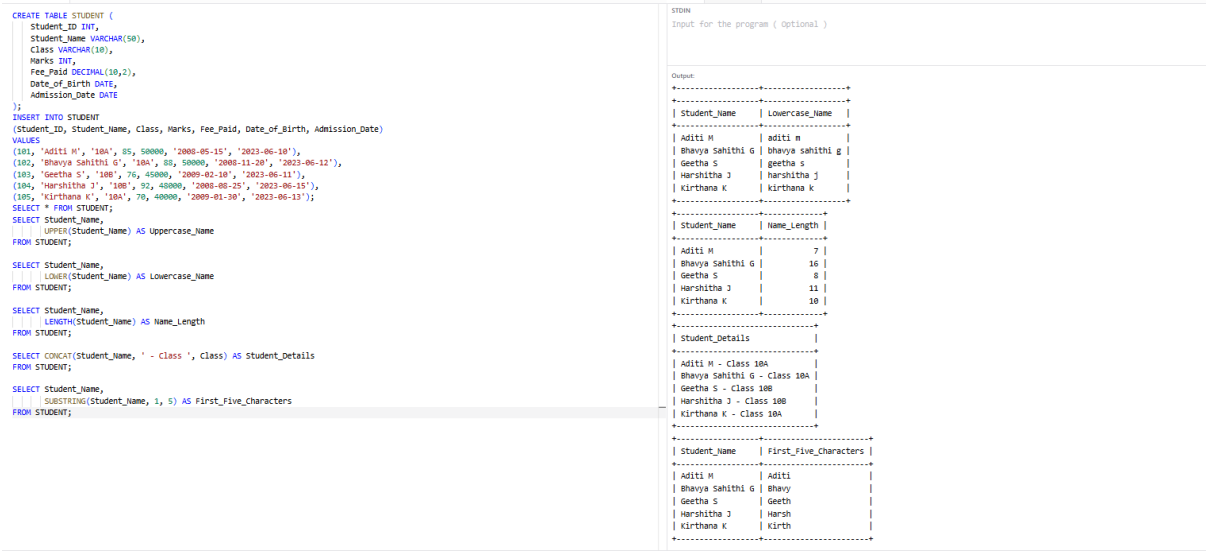
```
STDIN
Input for the program ( Optional )

Output:
+------------+-----------------+-------+----------+---------------+----------------+
| Student_ID | Student_Name    | Class | Marks    | Fee_Paid      | Date_of_Birth  | Admission_Date |
+------------+-----------------+-------+----------+---------------+----------------+
| 101        | Aditi M         | 10A   | 85       | 50000.00      | 2008-05-15     | 2023-06-10     |
| 102        | Bhavya Sahithi G| 10A   | 88       | 50000.00      | 2008-11-20     | 2023-06-12     |
| 103        | Geetha S        | 10B   | 76       | 45000.00      | 2009-02-10     | 2023-06-11     |
| 104        | Harshitha J     | 10B   | 92       | 48000.00      | 2008-08-25     | 2023-06-15     |
| 105        | Kirthana K      | 10A   | 70       | 40000.00      | 2009-01-30     | 2023-06-13     |
+------------+-----------------+-------+----------+---------------+----------------+
+-----------------+-----------------+
| Student_Name    | Uppercase_Name  |
+-----------------+-----------------+
| Aditi M         | ADITI M         |
| Bhavya Sahithi G| BHAVYA SAHITHI G|
| Geetha S        | GEETHA S        |
| Harshitha J     | HARSHITHA J     |
| Kirthana K      | KIRTHANA K      |
+-----------------+-----------------+
+-----------------+-----------------+
| Student_Name    | Lowercase_Name  |
+-----------------+-----------------+
| Aditi M         | aditi m         |
| Bhavya Sahithi G| bhavya sahithi g|
| Geetha S        | geetha s        |
| Harshitha J     | harshitha j     |
| Kirthana K      | kirthana k      |
+-----------------+-----------------+
+-----------------+-------------+
| Student_Name    | Name_Length |
+-----------------+-------------+
| Aditi M         |           7 |
| Bhavya Sahithi G |          16 |
| Geetha S        |           8 |
| Harshitha J     |          11 |
| Kirthana K      |          10 |
+-----------------+-------------+
+-----------------------------+
| Student_Details             |
+-----------------------------+
| Aditi M - Class 10A         |
| Bhavya Sahithi G - Class 10A |
| Geetha S - Class 10B        |
| Harshitha J - Class 10B     |
| Kirthana K - Class 10A      |
+-----------------------------+
```

SUBSTRING() FUNCTION

Purpose

The SUBSTRING() function extracts a specific part of a string.



```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;
```

```
STDIN
Input for the program ( Optional )

Output:
+-----------------+-----------------+
| Student_Name    | Lowercase_Name  |
+-----------------+-----------------+
| Aditi M         | aditi m         |
| Bhavya Sahithi G| bhavya sahithi g|
| Geetha S        | geetha s        |
| Harshitha J     | harshitha j     |
| Kirthana K      | kirthana k      |
+-----------------+-----------------+
+-----------------+-------------+
| Student_Name    | Name_Length |
+-----------------+-------------+
| Aditi M         |           7 |
| Bhavya Sahithi G |          16 |
| Geetha S        |           8 |
| Harshitha J     |          11 |
| Kirthana K      |          10 |
+-----------------+-------------+
+-----------------------------+
| Student_Details             |
+-----------------------------+
| Aditi M - Class 10A         |
| Bhavya Sahithi G - Class 10A |
| Geetha S - Class 10B        |
| Harshitha J - Class 10B     |
| Kirthana K - Class 10A      |
+-----------------------------+
+-----------------+----------------------+
| Student_Name    | First_Five_Characters |
+-----------------+----------------------+
| Aditi M         | Aditi                |
| Bhavya Sahithi G | Bhavy                |
| Geetha S        | Geeth                |
| Harshitha J     | Harsh                |
| Kirthana K      | Kirth                |
+-----------------+----------------------+
```

PART B – NUMERIC FUNCTIONS

Numeric functions perform calculations on numerical data such as Marks and Fee_Paid.

ROUND() FUNCTION

Purpose

The ROUND() function rounds a numerical value to the nearest integer.

```sql
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;
```

STDIN

Input for the program (Optional)

Output:

```
+-----------------+-----------------+
+-----------------+-------------+
| Student_Name    | Name_Length |
+-----------------+-------------+
| Aditi M         |           7 |
| Bhavya Sahithi G |          16 |
| Geetha S        |           8 |
| Harshitha J     |          11 |
| Kirthana K      |          10 |
+-----------------+-------------+
+-------------------------------+
| Student_Details               |
+-------------------------------+
| Aditi M - Class 10A           |
| Bhavya Sahithi G - Class 10A  |
| Geetha S - Class 10B          |
| Harshitha J - Class 10B       |
| Kirthana K - Class 10A        |
+-------------------------------+
+-----------------+-------------------------+
| Student_Name    | First_Five_Characters |
+-----------------+-------------------------+
| Aditi M         | Aditi                   |
| Bhavya Sahithi G | Bhavy                   |
| Geetha S        | Geeth                   |
| Harshitha J     | Harsh                   |
| Kirthana K      | Kirth                   |
+-----------------+-------------------------+
+-----------------+-------+-------------+
| Student_Name    | Marks | Bonus_Marks |
+-----------------+-------+-------------+
| Aditi M         |    85 |          89 |
| Bhavya Sahithi G |    88 |          92 |
| Geetha S        |    76 |          80 |
| Harshitha J     |    92 |          97 |
| Kirthana K      |    70 |          74 |
+-----------------+-------+-------------+
```

CEIL() FUNCTION

Purpose

The CEIL() function rounds a number upward to the next integer.

```sql
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;
```

STDIN

Input for the program (Optional)

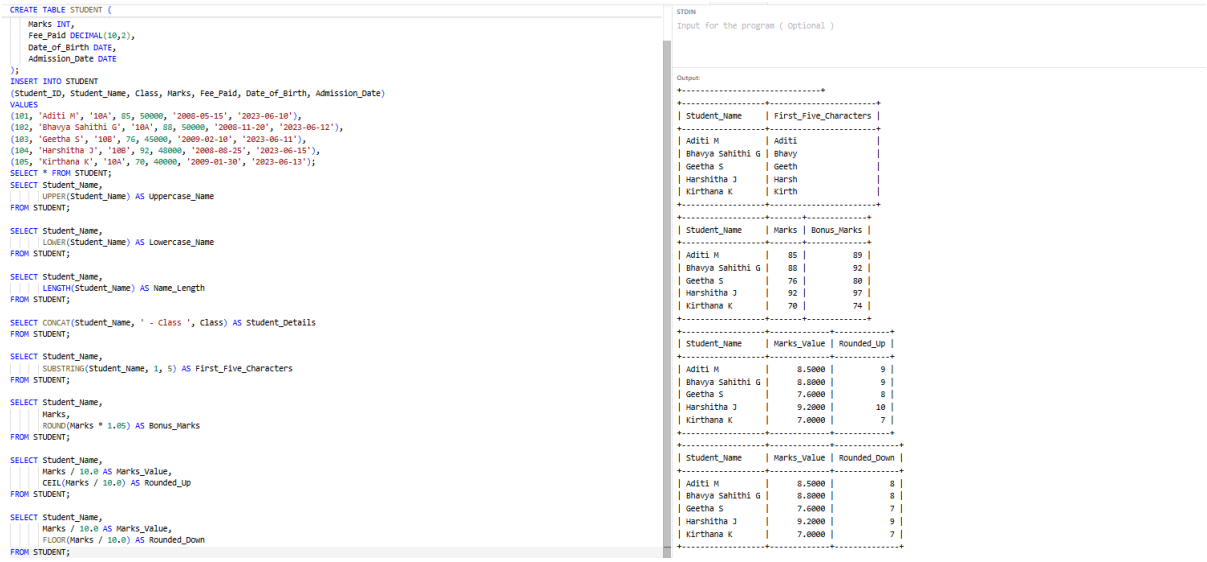
Output:

```
+-------------------------------+
| Student_Details               |
+-------------------------------+
| Aditi M - Class 10A           |
| Bhavya Sahithi G - Class 10A  |
| Geetha S - Class 10B          |
| Harshitha J - Class 10B       |
| Kirthana K - Class 10A        |
+-------------------------------+
+-----------------+-------------------------+
| Student_Name    | First_Five_Characters |
+-----------------+-------------------------+
| Aditi M         | Aditi                   |
| Bhavya Sahithi G | Bhavy                   |
| Geetha S        | Geeth                   |
| Harshitha J     | Harsh                   |
| Kirthana K      | Kirth                   |
+-----------------+-------------------------+
+-----------------+-------+-------------+
| Student_Name    | Marks | Bonus_Marks |
+-----------------+-------+-------------+
| Aditi M         |    85 |          89 |
| Bhavya Sahithi G |    88 |          92 |
| Geetha S        |    76 |          80 |
| Harshitha J     |    92 |          97 |
| Kirthana K      |    70 |          74 |
+-----------------+-------+-------------+
+-----------------+-------------+-------------+
| Student_Name    | Marks_Value | Rounded_Up |
+-----------------+-------------+-------------+
| Aditi M         |      8.5000 |          9 |
| Bhavya Sahithi G |      8.8000 |          9 |
| Geetha S        |      7.6000 |          8 |
| Harshitha J     |      9.2000 |         10 |
| Kirthana K      |      7.0000 |          7 |
+-----------------+-------------+-------------+
```

FLOOR() FUNCTION

Purpose

The FLOOR() function rounds a number downward to the nearest integer.



```
CREATE TABLE STUDENT (
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;
```

STDIN

Input for the program (Optional)

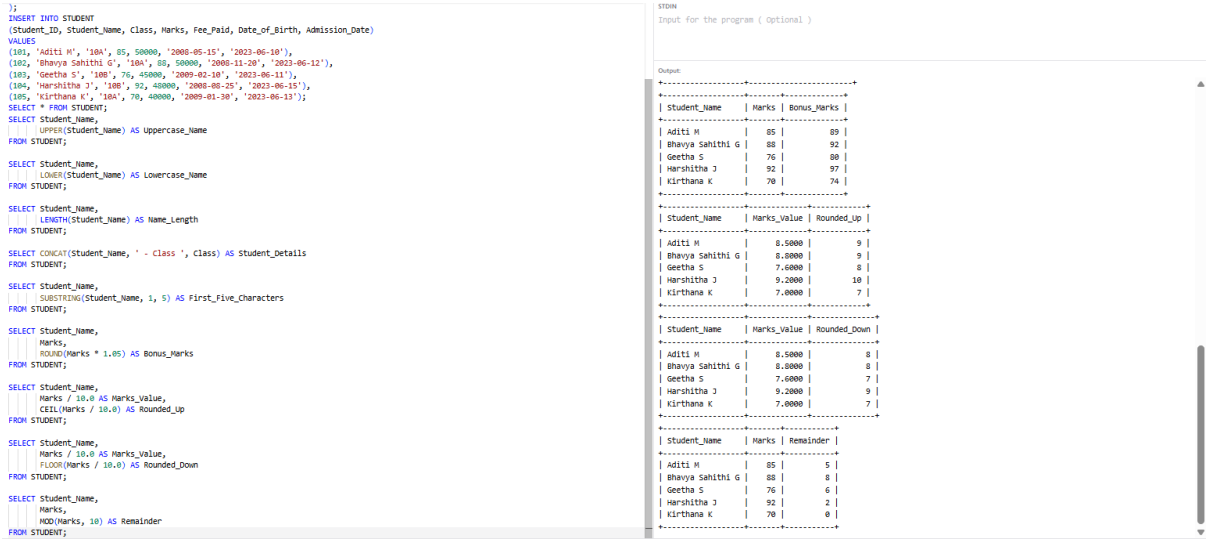
Output:

```
+-----------------------------+
+-----------------+-----------------------+
| Student_Name    | First_Five_Characters |
+-----------------+-----------------------+
| Aditi M         | Aditi                 |
| Bhavya Sahithi G | Bhavy                 |
| Geetha S        | Geeth                 |
| Harshitha J     | Harsh                 |
| Kirthana K      | Kirth                 |
+-----------------+-----------------------+
+-----------------+-------+-------------+
| Student_Name    | Marks | Bonus_Marks |
+-----------------+-------+-------------+
| Aditi M         |    85 |          89 |
| Bhavya Sahithi G |    88 |          92 |
| Geetha S        |    76 |          80 |
| Harshitha J     |    92 |          97 |
| Kirthana K      |    70 |          74 |
+-----------------+-------+-------------+
+-----------------+-------------+-------------+
| Student_Name    | Marks_Value | Rounded_Up  |
+-----------------+-------------+-------------+
| Aditi M         |      8.5000 |           9 |
| Bhavya Sahithi G |      8.8000 |           9 |
| Geetha S        |      7.6000 |           8 |
| Harshitha J     |      9.2000 |          10 |
| Kirthana K      |      7.0000 |           7 |
+-----------------+-------------+-------------+
+-----------------+-------------+--------------+
| Student_Name    | Marks_Value | Rounded_Down |
+-----------------+-------------+--------------+
| Aditi M         |      8.5000 |            8 |
| Bhavya Sahithi G |      8.8000 |            8 |
| Geetha S        |      7.6000 |            7 |
| Harshitha J     |      9.2000 |            9 |
| Kirthana K      |      7.0000 |            7 |
+-----------------+-------------+--------------+
```

MOD() FUNCTION

Purpose

The MOD() function returns the remainder after division.



```
);
INSERT INTO STUDENT
(Student_ID, Student_Name, Class, Marks, Fee_Paid, Date_of_Birth, Admission_Date)
VALUES
(101, 'Aditi M', '10A', 85, 50000, '2008-05-15', '2023-06-10'),
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;
```

STDIN

Input for the program (Optional)

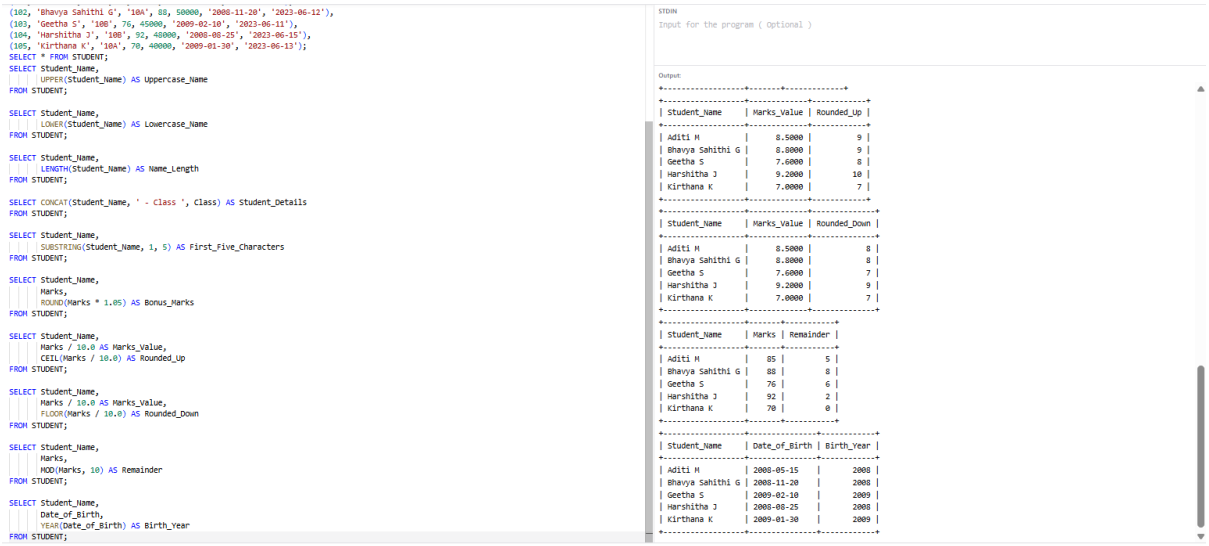
Output:

```
+-----------------+-------+-------------+
| Student_Name    | Marks | Bonus_Marks |
+-----------------+-------+-------------+
| Aditi M         |    85 |          89 |
| Bhavya Sahithi G |    88 |          92 |
| Geetha S        |    76 |          80 |
| Harshitha J     |    92 |          97 |
| Kirthana K      |    70 |          74 |
+-----------------+-------+-------------+
+-----------------+-------------+-------------+
| Student_Name    | Marks_Value | Rounded_Up  |
+-----------------+-------------+-------------+
| Aditi M         |      8.5000 |           9 |
| Bhavya Sahithi G |      8.8000 |           9 |
| Geetha S        |      7.6000 |           8 |
| Harshitha J     |      9.2000 |          10 |
| Kirthana K      |      7.0000 |           7 |
+-----------------+-------------+-------------+
+-----------------+-------------+--------------+
| Student_Name    | Marks_Value | Rounded_Down |
+-----------------+-------------+--------------+
| Aditi M         |      8.5000 |            8 |
| Bhavya Sahithi G |      8.8000 |            8 |
| Geetha S        |      7.6000 |            7 |
| Harshitha J     |      9.2000 |            9 |
| Kirthana K      |      7.0000 |            7 |
+-----------------+-------------+--------------+
+-----------------+-------+-----------+
| Student_Name    | Marks | Remainder |
+-----------------+-------+-----------+
| Aditi M         |    85 |         5 |
| Bhavya Sahithi G |    88 |         8 |
| Geetha S        |    76 |         6 |
| Harshitha J     |    92 |         2 |
| Kirthana K      |    70 |         0 |
+-----------------+-------+-----------+
```

ABS() FUNCTION

Purpose

The ABS() function returns the absolute value of a number. It can be used to find the difference between marks and a target value.



```
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;
```

```
STDIN
Input for the program ( Optional )

Output:
+-----------------+-------+------------+
+-----------------+-------+------------+
| Student_Name    | Marks_Value | Rounded_Up |
+-----------------+-------+------------+
| Aditi M         |      8.5000 |          9 |
| Bhavya Sahithi G |      8.8000 |          9 |
| Geetha S        |      7.6000 |          8 |
| Harshitha J     |      9.2000 |         10 |
| Kirthana K      |      7.0000 |          7 |
+-----------------+-------+------------+
+-----------------+-------+------------+
| Student_Name    | Marks_Value | Rounded_Down |
+-----------------+-------+------------+
| Aditi M         |      8.5000 |          8 |
| Bhavya Sahithi G |      8.8000 |          8 |
| Geetha S        |      7.6000 |          7 |
| Harshitha J     |      9.2000 |          9 |
| Kirthana K      |      7.0000 |          7 |
+-----------------+-------+------------+
+-----------------+-------+-----------+
| Student_Name    | Marks | Remainder |
+-----------------+-------+-----------+
| Aditi M         |    85 |         5 |
| Bhavya Sahithi G |    88 |         8 |
| Geetha S        |    76 |         6 |
| Harshitha J     |    92 |         2 |
| Kirthana K      |    70 |         0 |
+-----------------+-------+-----------+
+-----------------+---------------+------------+
| Student_Name    | Date_of_Birth | Birth_Year |
+-----------------+---------------+------------+
| Aditi M         | 2008-05-15    |       2008 |
| Bhavya Sahithi G | 2008-11-20    |       2008 |
| Geetha S        | 2009-02-10    |       2009 |
| Harshitha J     | 2008-08-25    |       2008 |
| Kirthana K      | 2009-01-30    |       2009 |
+-----------------+---------------+------------+
```

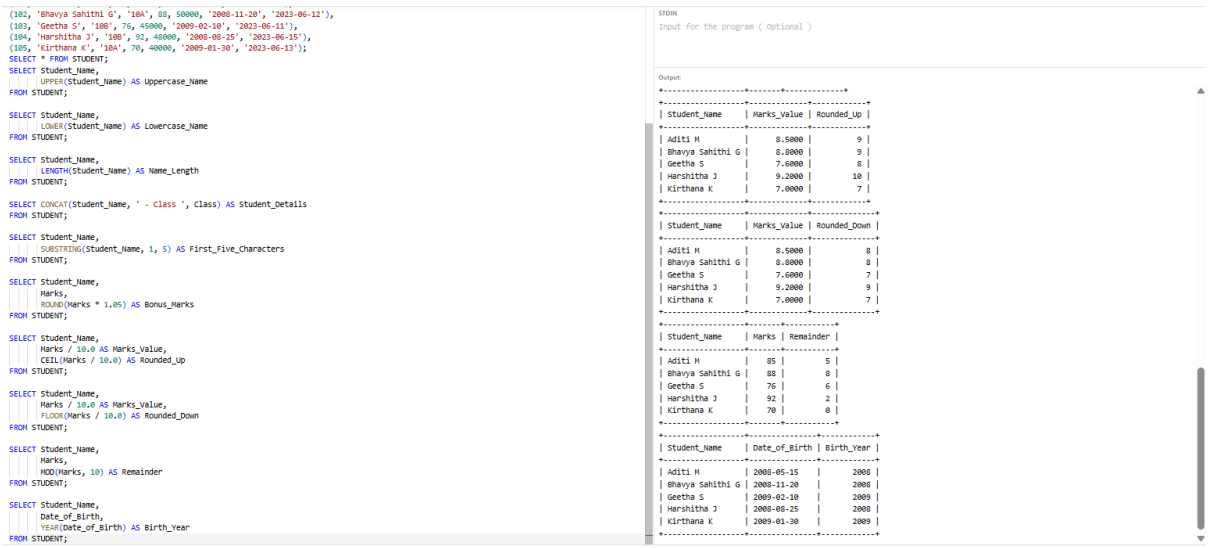
PART C – DATE FUNCTIONS

Date functions are used to work with date values such as Date of Birth and Admission Date.

YEAR() FUNCTION

Purpose

The YEAR() function extracts the year from a date.



```
(102, 'Bhavya Sahithi G', '10A', 88, 50000, '2008-11-20', '2023-06-12'),
(103, 'Geetha S', '10B', 76, 45000, '2009-02-10', '2023-06-11'),
(104, 'Harshitha J', '10B', 92, 48000, '2008-08-25', '2023-06-15'),
(105, 'Kirthana K', '10A', 70, 40000, '2009-01-30', '2023-06-13');
SELECT * FROM STUDENT;
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;
```

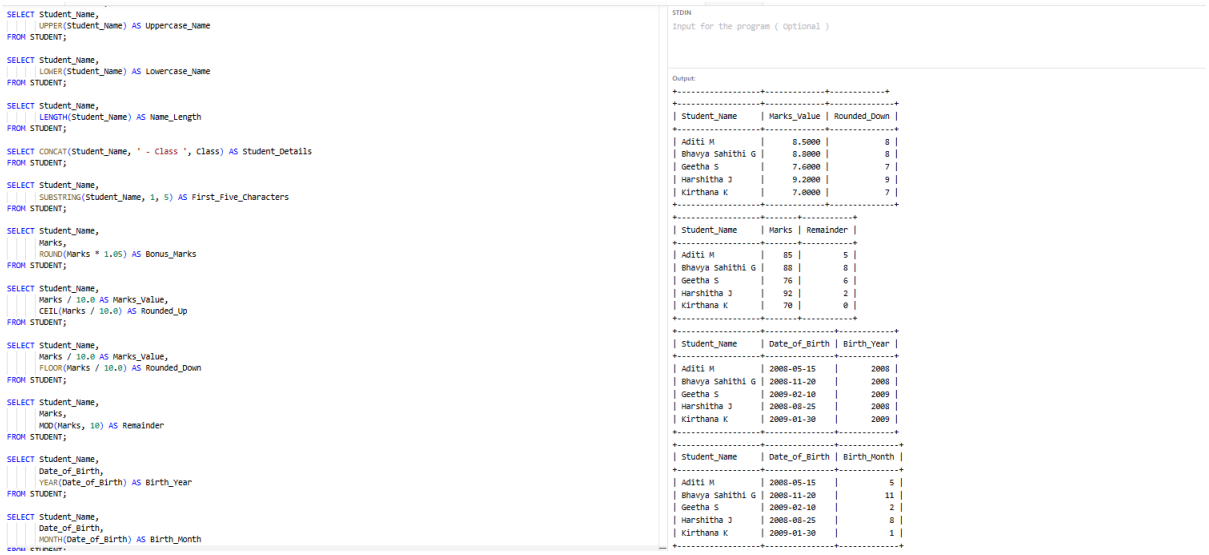
```
STDIN
Input for the program ( Optional )

Output:
+-----------------+-------+------------+
+-----------------+-------+------------+
| Student_Name    | Marks_Value | Rounded_Up |
+-----------------+-------+------------+
| Aditi M         |      8.5000 |          9 |
| Bhavya Sahithi G |      8.8000 |          9 |
| Geetha S        |      7.6000 |          8 |
| Harshitha J     |      9.2000 |         10 |
| Kirthana K      |      7.0000 |          7 |
+-----------------+-------+------------+
+-----------------+-------+------------+
| Student_Name    | Marks_Value | Rounded_Down |
+-----------------+-------+------------+
| Aditi M         |      8.5000 |          8 |
| Bhavya Sahithi G |      8.8000 |          8 |
| Geetha S        |      7.6000 |          7 |
| Harshitha J     |      9.2000 |          9 |
| Kirthana K      |      7.0000 |          7 |
+-----------------+-------+------------+
+-----------------+-------+-----------+
| Student_Name    | Marks | Remainder |
+-----------------+-------+-----------+
| Aditi M         |    85 |         5 |
| Bhavya Sahithi G |    88 |         8 |
| Geetha S        |    76 |         6 |
| Harshitha J     |    92 |         2 |
| Kirthana K      |    70 |         0 |
+-----------------+-------+-----------+
+-----------------+---------------+------------+
| Student_Name    | Date_of_Birth | Birth_Year |
+-----------------+---------------+------------+
| Aditi M         | 2008-05-15    |       2008 |
| Bhavya Sahithi G | 2008-11-20    |       2008 |
| Geetha S        | 2009-02-10    |       2009 |
| Harshitha J     | 2008-08-25    |       2008 |
| Kirthana K      | 2009-01-30    |       2009 |
+-----------------+---------------+------------+
```

MONTH() FUNCTION

Purpose

The MONTH() function extracts the month from a date.



```
SELECT Student_Name,
       UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;
```

STDIN

Input for the program (Optional)

Output:

Student_Name	Marks_Value	Rounded_Down
Aditi M	8.5000	8
Bhavya Sahithi G	8.8000	8
Geetha S	7.6000	7
Harshitha J	9.2000	9
Kirthana K	7.0000	7

Student_Name	Marks	Remainder
Aditi M	85	5
Bhavya Sahithi G	88	8
Geetha S	76	6
Harshitha J	92	2
Kirthana K	70	0

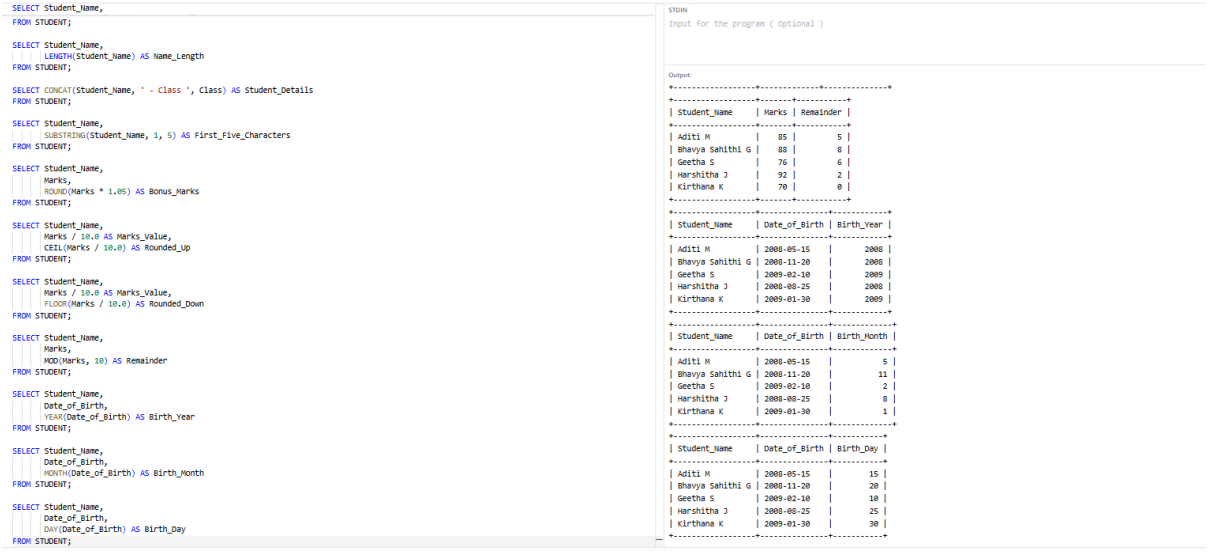
Student_Name	Date_of_Birth	Birth_Year
Aditi M	2008-05-15	2008
Bhavya Sahithi G	2008-11-20	2008
Geetha S	2009-02-10	2009
Harshitha J	2008-08-25	2008
Kirthana K	2009-01-30	2009

Student_Name	Date_of_Birth	Birth_Month
Aditi M	2008-05-15	5
Bhavya Sahithi G	2008-11-20	11
Geetha S	2009-02-10	2
Harshitha J	2008-08-25	8
Kirthana K	2009-01-30	1

DAY() FUNCTION

Purpose

The DAY() function extracts the day from a date.



```
SELECT Student_Name,
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
```

STDIN

Input for the program (Optional)

Output:

Student_Name	Marks	Remainder
Aditi M	85	5
Bhavya Sahithi G	88	8
Geetha S	76	6
Harshitha J	92	2
Kirthana K	70	0

Student_Name	Date_of_Birth	Birth_Year
Aditi M	2008-05-15	2008
Bhavya Sahithi G	2008-11-20	2008
Geetha S	2009-02-10	2009
Harshitha J	2008-08-25	2008
Kirthana K	2009-01-30	2009

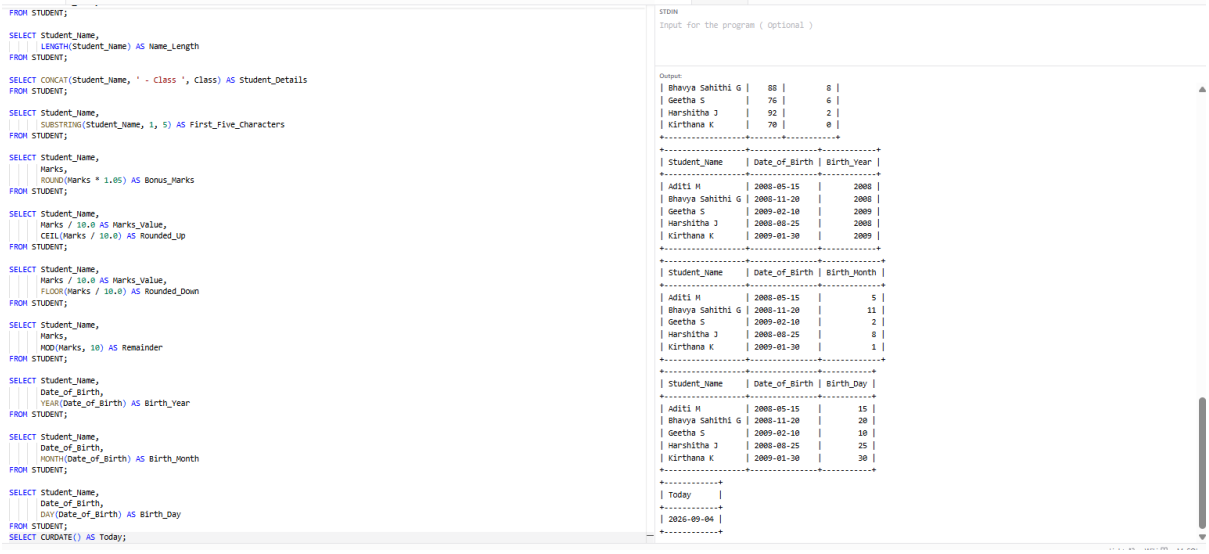
Student_Name	Date_of_Birth	Birth_Month
Aditi M	2008-05-15	5
Bhavya Sahithi G	2008-11-20	11
Geetha S	2009-02-10	2
Harshitha J	2008-08-25	8
Kirthana K	2009-01-30	1

Student_Name	Date_of_Birth	Birth_Day
Aditi M	2008-05-15	15
Bhavya Sahithi G	2008-11-20	20
Geetha S	2009-02-10	10
Harshitha J	2008-08-25	25
Kirthana K	2009-01-30	30

TIMESTAMPDIFF() FUNCTION

Purpose

The TIMESTAMPDIFF() function calculates the difference between two dates. It can be used to calculate the age of students.



```
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_Of_Birth,
       YEAR(Date_Of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_Of_Birth,
       MONTH(Date_Of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_Of_Birth,
       DAY(Date_Of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;
```

```
STDIN
Input for the program ( Optional )

Output:
| Bhavya Sahithi G |    88 |        8 |
| Geetha S         |    76 |        6 |
| Harshitha J      |    92 |        2 |
| Kirthana K       |    70 |        0 |
+------------------+-------+----------+

+------------------+---------------+------------+
| Student_Name     | Date_of_Birth | Birth_Year |
+------------------+---------------+------------+
| Aditi M          | 2008-05-15    |       2008 |
| Bhavya Sahithi G | 2008-11-20    |       2008 |
| Geetha S         | 2009-02-10    |       2009 |
| Harshitha J      | 2008-08-25    |       2008 |
| Kirthana K       | 2009-01-30    |       2009 |
+------------------+---------------+------------+

+------------------+---------------+-------------+
| Student_Name     | Date_of_Birth | Birth_Month |
+------------------+---------------+-------------+
| Aditi M          | 2008-05-15    |           5 |
| Bhavya Sahithi G | 2008-11-20    |          11 |
| Geetha S         | 2009-02-10    |           2 |
| Harshitha J      | 2008-08-25    |           8 |
| Kirthana K       | 2009-01-30    |           1 |
+------------------+---------------+-------------+

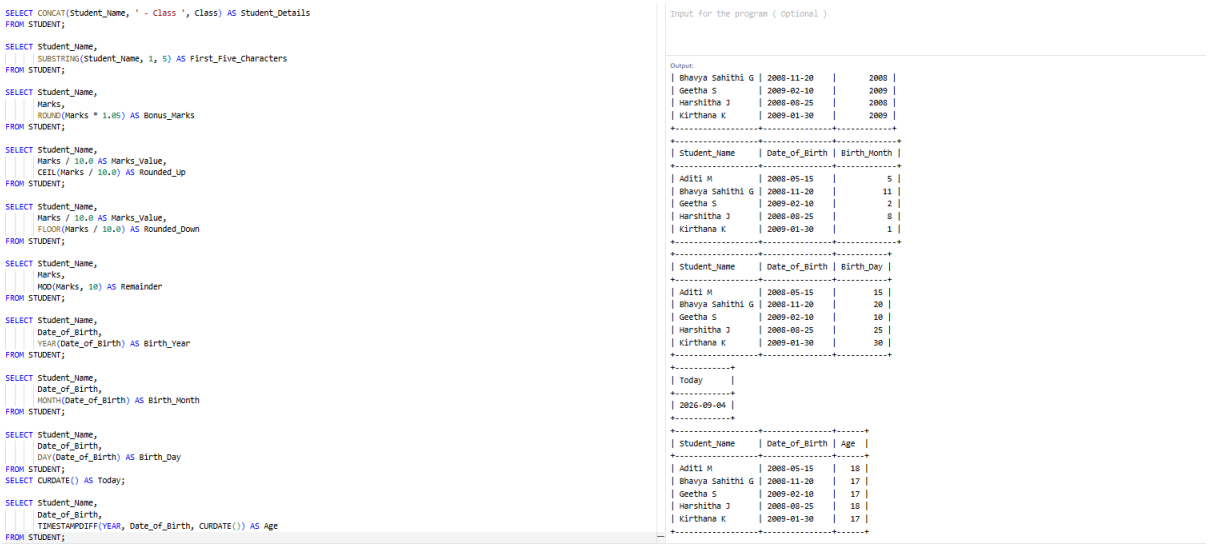
+------------------+---------------+-----------+
| Student_Name     | Date_of_Birth | Birth_Day |
+------------------+---------------+-----------+
| Aditi M          | 2008-05-15    |        15 |
| Bhavya Sahithi G | 2008-11-20    |        20 |
| Geetha S         | 2009-02-10    |        10 |
| Harshitha J      | 2008-08-25    |        25 |
| Kirthana K       | 2009-01-30    |        30 |
+------------------+---------------+-----------+

+------------+
| Today      |
+------------+
| 2026-09-04 |
+------------+
```

CURDATE() FUNCTION

Purpose

The CURDATE() function returns the current date.



```
SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_Of_Birth,
       YEAR(Date_Of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_Of_Birth,
       MONTH(Date_Of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_Of_Birth,
       DAY(Date_Of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_Of_Birth,
       TIMESTAMPDIFF(YEAR, Date_Of_Birth, CURDATE()) AS Age
FROM STUDENT;
```

```
Input for the program ( Optional )

Output:
| Bhavya Sahithi G | 2008-11-20    |       2008 |
| Geetha S         | 2009-02-10    |       2009 |
| Harshitha J      | 2008-08-25    |       2008 |
| Kirthana K       | 2009-01-30    |       2009 |
+------------------+---------------+------------+

+------------------+---------------+-------------+
| Student_Name     | Date_Of_Birth | Birth_Month |
+------------------+---------------+-------------+
| Aditi M          | 2008-05-15    |           5 |
| Bhavya Sahithi G | 2008-11-20    |          11 |
| Geetha S         | 2009-02-10    |           2 |
| Harshitha J      | 2008-08-25    |           8 |
| Kirthana K       | 2009-01-30    |           1 |
+------------------+---------------+-------------+

+------------------+---------------+-----------+
| Student_Name     | Date_Of_Birth | Birth_Day |
+------------------+---------------+-----------+
| Aditi M          | 2008-05-15    |        15 |
| Bhavya Sahithi G | 2008-11-20    |        20 |
| Geetha S         | 2009-02-10    |        10 |
| Harshitha J      | 2008-08-25    |        25 |
| Kirthana K       | 2009-01-30    |        30 |
+------------------+---------------+-----------+

+------------+
| Today      |
+------------+
| 2026-09-04 |
+------------+

+------------------+---------------+------+
| Student_Name     | Date_Of_Birth | Age  |
+------------------+---------------+------+
| Aditi M          | 2008-05-15    |   18 |
| Bhavya Sahithi G | 2008-11-20    |   17 |
| Geetha S         | 2009-02-10    |   17 |
| Harshitha J      | 2008-08-25    |   18 |
| Kirthana K       | 2009-01-30    |   17 |
+------------------+---------------+------+
```

PART D – AGGREGATE FUNCTIONS

Aggregate functions operate on multiple rows together and produce a single result.

COUNT() FUNCTION

Purpose

The COUNT() function counts the number of records.

```sql
SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;
```

```
Input for the program ( Optional )

Output:
| Bhavya Sahithi G | 2008-11-20    |       2008 |
| Geetha S         | 2009-02-10    |       2009 |
| Harshitha J      | 2008-08-25    |       2008 |
| Kirthana K       | 2009-01-30    |       2009 |
+------------------+---------------+------------+
+------------------+---------------+------------+
| Student_Name     | Date_of_Birth | Birth_Month |
+------------------+---------------+------------+
| Aditi M          | 2008-05-15    |           5 |
| Bhavya Sahithi G | 2008-11-20    |          11 |
| Geetha S         | 2009-02-10    |           2 |
| Harshitha J      | 2008-08-25    |           8 |
| Kirthana K       | 2009-01-30    |           1 |
+------------------+---------------+------------+
+------------------+---------------+----------+
| Student_Name     | Date_of_Birth | Birth_Day |
+------------------+---------------+----------+
| Aditi M          | 2008-05-15    |        15 |
| Bhavya Sahithi G | 2008-11-20    |        20 |
| Geetha S         | 2009-02-10    |        10 |
| Harshitha J      | 2008-08-25    |        25 |
| Kirthana K       | 2009-01-30    |        30 |
+------------------+---------------+----------+
+------------+
| Today      |
+------------+
| 2026-09-04 |
+------------+
+------------------+---------------+-----+
| Student_Name     | Date_of_Birth | Age |
+------------------+---------------+-----+
| Aditi M          | 2008-05-15    |  18 |
| Bhavya Sahithi G | 2008-11-20    |  17 |
| Geetha S         | 2009-02-10    |  17 |
| Harshitha J      | 2008-08-25    |  18 |
| Kirthana K       | 2009-01-30    |  17 |
+------------------+---------------+-----+
```

SUM() FUNCTION

Purpose

The SUM() function calculates the total of numerical values.

```sql
SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;
```

```
STDIN
Input for the program ( Optional )

Output:
+------------------+---------------+------------+
| Student_Name     | Date_of_Birth | Birth_Month |
+------------------+---------------+------------+
| Aditi M          | 2008-05-15    |           5 |
| Bhavya Sahithi G | 2008-11-20    |          11 |
| Geetha S         | 2009-02-10    |           2 |
| Harshitha J      | 2008-08-25    |           8 |
| Kirthana K       | 2009-01-30    |           1 |
+------------------+---------------+------------+
+------------------+---------------+----------+
| Student_Name     | Date_of_Birth | Birth_Day |
+------------------+---------------+----------+
| Aditi M          | 2008-05-15    |        15 |
| Bhavya Sahithi G | 2008-11-20    |        20 |
| Geetha S         | 2009-02-10    |        10 |
| Harshitha J      | 2008-08-25    |        25 |
| Kirthana K       | 2009-01-30    |        30 |
+------------------+---------------+----------+
+------------+
| Today      |
+------------+
| 2026-09-04 |
+------------+
+------------------+---------------+-----+
| Student_Name     | Date_of_Birth | Age |
+------------------+---------------+-----+
| Aditi M          | 2008-05-15    |  18 |
| Bhavya Sahithi G | 2008-11-20    |  17 |
| Geetha S         | 2009-02-10    |  17 |
| Harshitha J      | 2008-08-25    |  18 |
| Kirthana K       | 2009-01-30    |  17 |
+------------------+---------------+-----+
+---------------+
| Total_Students |
+---------------+
|             5 |
+---------------+
```

AVG() FUNCTION

Purpose

The AVG() function calculates the average of numerical values.

```sql
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;
```

STDIN

Input for the program (Optional)

Output:

```
| Geetha S         | 2009-02-10    |           2 |
| Harshitha J      | 2008-08-25    |           8 |
| Kirthana K       | 2009-01-30    |           1 |
+------------------+---------------+-------------+
+------------------+---------------+-----------+
| Student_Name     | Date_of_Birth | Birth_Day |
+------------------+---------------+-----------+
| Aditi M          | 2008-05-15    |        15 |
| Bhavya Sahithi G | 2008-11-20    |        20 |
| Geetha S         | 2009-02-10    |        10 |
| Harshitha J      | 2008-08-25    |        25 |
| Kirthana K       | 2009-01-30    |        30 |
+------------------+---------------+-----------+
+------------+
| Today      |
+------------+
| 2026-09-04 |
+------------+
+------------------+---------------+------+
| Student_Name     | Date_of_Birth | Age  |
+------------------+---------------+------+
| Aditi M          | 2008-05-15    |   18 |
| Bhavya Sahithi G | 2008-11-20    |   17 |
| Geetha S         | 2009-02-10    |   17 |
| Harshitha J      | 2008-08-25    |   18 |
| Kirthana K       | 2009-01-30    |   17 |
+------------------+---------------+------+
+---------------+
| Total_Students |
+---------------+
|             5 |
+---------------+
+-------------+
| Total_Marks |
+-------------+
|         411 |
+-------------+
```

MAX() FUNCTION

Purpose

The MAX() function finds the highest value from a column.

```sql
SELECT Student_Name,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;
```

STDIN

Input for the program (Optional)

Output:

```
| Student_Name     | Date_of_Birth | Birth_Day |
+------------------+---------------+-----------+
| Aditi M          | 2008-05-15    |        15 |
| Bhavya Sahithi G | 2008-11-20    |        20 |
| Geetha S         | 2009-02-10    |        10 |
| Harshitha J      | 2008-08-25    |        25 |
| Kirthana K       | 2009-01-30    |        30 |
+------------------+---------------+-----------+
+------------+
| Today      |
+------------+
| 2026-09-04 |
+------------+
+------------------+---------------+------+
| Student_Name     | Date_of_Birth | Age  |
+------------------+---------------+------+
| Aditi M          | 2008-05-15    |   18 |
| Bhavya Sahithi G | 2008-11-20    |   17 |
| Geetha S         | 2009-02-10    |   17 |
| Harshitha J      | 2008-08-25    |   18 |
| Kirthana K       | 2009-01-30    |   17 |
+------------------+---------------+------+
+---------------+
| Total_Students |
+---------------+
|             5 |
+---------------+
+-------------+
| Total_Marks |
+-------------+
|         411 |
+-------------+
+---------------+
| Average_Marks |
+---------------+
|       82.2000 |
+---------------+
```

MIN() FUNCTION

Purpose

The MIN() function finds the lowest value from a column.



```
SELECT Student_Name,
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;

SELECT MAX(Marks) AS Highest_Marks
FROM STUDENT;

SELECT MIN(Marks) AS Lowest_Marks
FROM STUDENT;
```

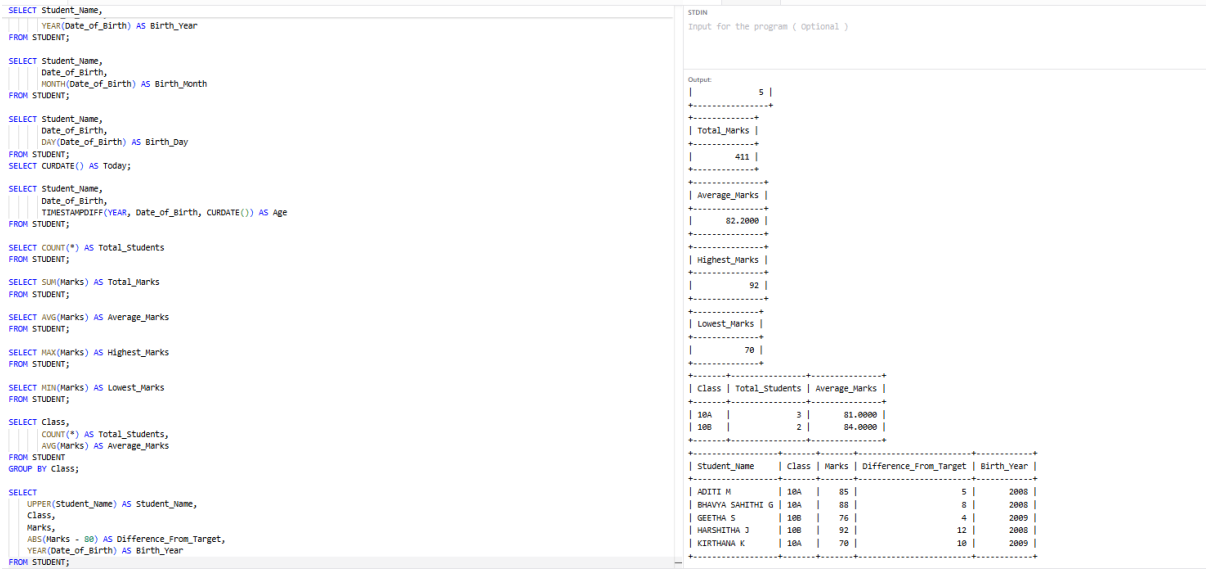
```
STDIN
Input for the program ( Optional )

Output:
+------------+
| 2026-09-04 |
+------------+
+------------------+---------------+------+
| Student_Name     | Date_of_Birth | Age  |
+------------------+---------------+------+
| Aditi M          | 2008-05-15    |   18 |
| Bhavya Sahithi G | 2008-11-20    |   17 |
| Geetha S         | 2009-02-10    |   17 |
| Harshitha J      | 2008-08-25    |   18 |
| Kirthana K       | 2009-01-30    |   17 |
+------------------+---------------+------+
+---------------+
| Total_Students |
+---------------+
|              5 |
+---------------+
+-------------+
| Total_Marks |
+-------------+
|         411 |
+-------------+
+---------------+
| Average_Marks |
+---------------+
|       82.2000 |
+---------------+
+---------------+
| Highest_Marks |
+---------------+
|            92 |
+---------------+
+-------------+
| Lowest_Marks |
+-------------+
|          70 |
+-------------+
```

AGGREGATE FUNCTIONS USING GROUP BY

Purpose

The GROUP BY clause is used to group records based on a column. Here, students are grouped according to their class, and the number of students and average marks are calculated for each class.



```
SELECT Student_Name,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;

SELECT MAX(Marks) AS Highest_Marks
FROM STUDENT;

SELECT MIN(Marks) AS Lowest_Marks
FROM STUDENT;

SELECT Class,
       COUNT(*) AS Total_Students,
       AVG(Marks) AS Average_Marks
FROM STUDENT
GROUP BY Class;

SELECT
    UPPER(Student_Name) AS Student_Name,
    Class,
    Marks,
    ABS(Marks - 80) AS Difference_From_Target,
    YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;
```

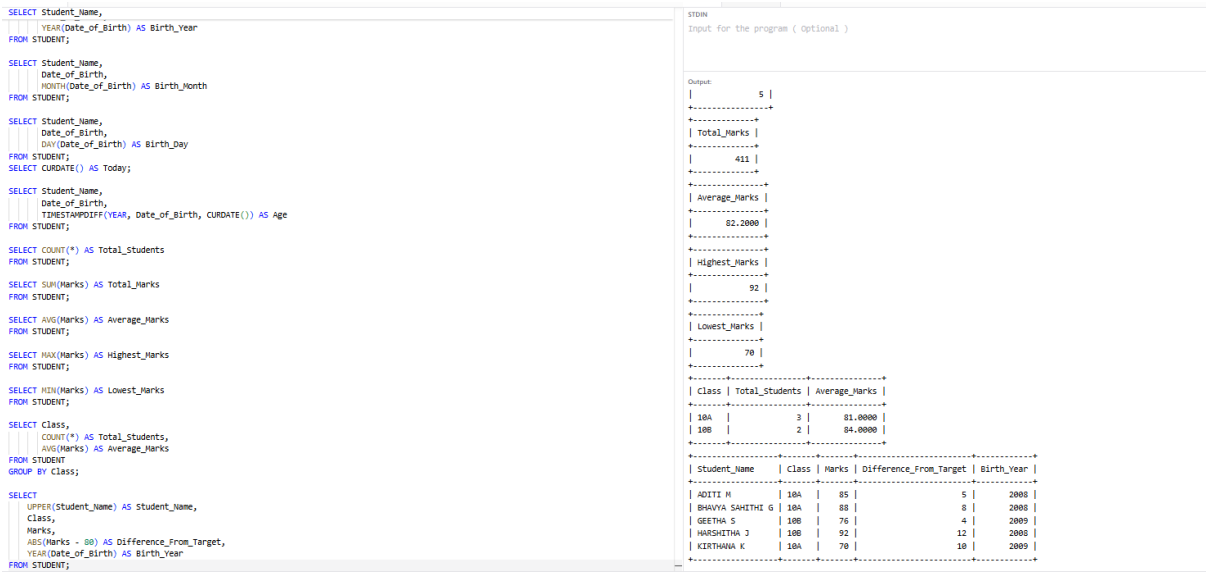
```
STDIN
Input for the program ( Optional )

Output:
|              5 |
+----------------+
+-------------+
| Total_Marks |
+-------------+
|         411 |
+-------------+
+---------------+
| Average_Marks |
+---------------+
|       82.2000 |
+---------------+
+---------------+
| Highest_Marks |
+---------------+
|            92 |
+---------------+
+-------------+
| Lowest_Marks |
+-------------+
|          70 |
+-------------+
+-------+----------------+---------------+
| Class | Total_Students | Average_Marks |
+-------+----------------+---------------+
| 10A   |              3 |       81.0000 |
| 10B   |              2 |       84.0000 |
+-------+----------------+---------------+
+------------------+-------+-------+--------------------------+------------+
| Student_Name     | Class | Marks | Difference_From_Target | Birth_Year |
+------------------+-------+-------+--------------------------+------------+
| ADITI M          | 10A   |    85 |                        5 |       2008 |
| BHAVYA SAHITHI G | 10A   |    88 |                        8 |       2008 |
| GEETHA S         | 10B   |    76 |                        4 |       2009 |
| HARSHITHA J      | 10B   |    92 |                       12 |       2008 |
| KIRTHANA K       | 10A   |    70 |                       10 |       2009 |
+------------------+-------+-------+--------------------------+------------+
```

COMBINED SQL FUNCTIONS

Purpose

This query combines multiple SQL functions to generate a student report containing the student's name in uppercase, class, marks, difference from the target mark of 80, and birth year.



```
SELECT Student_Name,
       YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
       Date_of_Birth,
       DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;
SELECT CURDATE() AS Today;

SELECT Student_Name,
       Date_of_Birth,
       TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;

SELECT MAX(Marks) AS Highest_Marks
FROM STUDENT;

SELECT MIN(Marks) AS Lowest_Marks
FROM STUDENT;

SELECT Class,
       COUNT(*) AS Total_Students,
       AVG(Marks) AS Average_Marks
FROM STUDENT
GROUP BY Class;

SELECT
    UPPER(Student_Name) AS Student_Name,
    Class,
    Marks,
    ABS(Marks - 80) AS Difference_From_Target,
    YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;
```

STDIN

Input for the program (Optional)

Output:

```
|             5 |
+---------------+
+-------------+
| Total_Marks |
+-------------+
|         411 |
+-------------+
+---------------+
| Average_Marks |
+---------------+
|       82.2000 |
+---------------+
+---------------+
| Highest_Marks |
+---------------+
|            92 |
+---------------+
+-------------+
| Lowest_Marks |
+-------------+
|          70 |
+-------------+
+-------+----------------+---------------+
| Class | Total_Students | Average_Marks |
+-------+----------------+---------------+
| 10A   |              3 |       81.0000 |
| 10B   |              2 |       84.0000 |
+-------+----------------+---------------+
+-----------------+-------+-------+-------------------------+------------+
| Student_Name    | Class | Marks | Difference_From_Target | Birth_Year |
+-----------------+-------+-------+-------------------------+------------+
| ADITI M         | 10A   |    85 |                       5 |       2008 |
| BHAVYA SAHITHI G | 10A   |    88 |                       8 |       2008 |
| GEETHA S        | 10B   |    76 |                       4 |       2009 |
| HARSHITHA J     | 10B   |    92 |                      12 |       2008 |
| KIRTHANA K      | 10A   |    70 |                      10 |       2009 |
+-----------------+-------+-------+-------------------------+------------+
```

CONCLUSION

The SQL functions were successfully applied to the STUDENT database. String functions were used to manipulate text data, Numeric functions were used to perform calculations, Date functions were used to extract and calculate date-related information, and Aggregate functions were used to perform calculations on multiple records. The GROUP BY clause was also used to obtain class-wise results, and a combined query was executed using multiple SQL functions.